

The exponential rate for four-term-progression-free subset-sum sets

Anonymous/deleted Reddit contributor
proof construction
The Clankers
independent reconstruction, verification, and formalization

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Abstract

For a finite set A of positive integers, let $H(A)$ be its set of subset sums, including zero, and let $g_4(n)$ be the least N for which an n -element set $A \subseteq \{1, \dots, N\}$ has $H(A)$ free of nonconstant four-term arithmetic progressions. We give a complete reconstruction and independent check of a proof posted by a now-deleted Reddit account. The proof determines the exponential rate:

$$\lim_{n \rightarrow \infty} g_4(n)^{1/n} = 19^{1/3}.$$

The lower bound classifies every relation with coefficients in $\{-2, -1, 0, 1, 2\}$ into disjoint minimal signed-relation blocks and counts the resulting ternary collision classes. The upper bound uses the block $\{1, 7, 8\}$, whose seven subset-sum residues form a four-progression-free digit set modulo 19, and lifts it without carries in base 19. We state every compressed step explicitly, record small typographical corrections to the source, and supply Lean and executable certificates whose exact scopes are listed in the final section.

Attribution and claim of credit. The construction and proof route in this note are attributed to the anonymous contributor who posted the source on r/LLMmathematics under an account that was subsequently deleted [1]. The linked ChatGPT conversation is also retained as a public source locator [2]. The Clankers do not claim priority for that construction; “The Clankers” denotes the independent reconstruction, checking, exposition, and formalization recorded here.

The moderators retain contemporaneous private mod-mail or notification evidence of the original Reddit username. The contributor is invited to request any public attribution they prefer; the private record can be used to validate a claimant without publishing the deleted username. In short, the proof is theirs, and this note is intended to make it checkable and return it to them with its provenance intact.

1 Problem and result

For a finite set $A = \{a_1, \dots, a_n\}$ of distinct positive integers, define

$$H(A) = \left\{ \sum_{i=1}^n \varepsilon_i a_i : \varepsilon_i \in \{0, 1\} \right\}.$$

A four-term arithmetic progression is an ordered tuple $(x, x + d, x + 2d, x + 3d)$; it is nonconstant when $d \neq 0$. Define $g_4(n)$ to be the least positive integer N for which there is an n -element set $A \subseteq \{1, \dots, N\}$ such that $H(A)$ contains no nonconstant four-term arithmetic progression.

This is the $k = 4$ exponential-rate subproblem of Erdős Problem 817, which originates in work of Erdős and Sárközy [5, 3]. Korsky's recent general results give, for fixed $k \geq 4$, a lower exponential base $(k - 1)/(k - 2)$ and a separate restricted-digit upper construction [6, Theorems 1.2 and 1.3]. The result below specializes to $k = 4$ but obtains the exact exponential rate. It does not address all of Problem 817. In particular, it is logically separate from the highlighted $k = 3$ question, to which Costa has recently given a negative answer by proving $\liminf_{n \rightarrow \infty} g_3(n)/3^n = 0$ [4].

Theorem 1.1 (Main theorem). *For every positive integer n ,*

$$\frac{19^{n/3} - 1}{2n} \leq g_4(n) \leq 8 \cdot 19^{\lceil n/3 \rceil - 1}.$$

Consequently,

$$\lim_{n \rightarrow \infty} g_4(n)^{1/n} = 19^{1/3}.$$

2 Relation splitting

Assume throughout Sections 2.1–3 that $A = \{a_1, \dots, a_n\} \subseteq \{1, \dots, N\}$ and that $H(A)$ is free of nonconstant four-term arithmetic progressions.

2.1 The two coefficient layers

Lemma 2.1 (Relation splitting). *Suppose*

$$\sum_{i=1}^n c_i a_i = 0, \quad c_i \in \{-2, -1, 0, 1, 2\}.$$

Define

$$P_j = \sum_{c_i=j} a_i, \quad N_j = \sum_{c_i=-j} a_i \quad (j = 1, 2).$$

Then $P_1 = N_1$ and $P_2 = N_2$.

Proof. The relation is

$$P_1 + 2P_2 = N_1 + 2N_2.$$

Set

$$X_0 = P_1 + N_2, \quad X_1 = P_1 + P_2, \quad X_2 = N_1 + N_2, \quad X_3 = N_1 + P_2.$$

The four coefficient classes are disjoint, so every X_j is a subset sum of A . Put $d = P_2 - N_2$. The displayed relation gives $N_1 - P_1 = 2d$, and direct subtraction, without changing coordinates, gives

$$X_1 - X_0 = d,$$

$$X_2 - X_1 = (N_1 - P_1) + (N_2 - P_2) = 2d - d = d,$$

and

$$X_3 - X_2 = P_2 - N_2 = d.$$

Thus (X_0, X_1, X_2, X_3) is a four-term arithmetic progression in $H(A)$. It must be constant, so $d = 0$. Hence $P_2 = N_2$, and the original relation then gives $P_1 = N_1$. $\square$

Equivalently, if a coefficient vector in $[-2, 2]^n \cap \mathbb{Z}^n$ is a relation, then its magnitude-one layer and its magnitude-two layer, after division by two, are separately signed relations.

2.2 Minimal signed-relation blocks

A *signed relation* is a nonzero vector $r \in \{-1, 0, 1\}^n$ satisfying $\sum_i r_i a_i = 0$. It is *minimal* if no nonzero signed relation has support properly contained in $\text{supp } r$.

Lemma 2.2 (Minimal blocks are disjoint). *If r and s are minimal signed relations, then either $\text{supp } r \cap \text{supp } s = \emptyset$ or $s = \pm r$.*

Proof. First let r be minimal and let s be any signed relation. Define

$$E_+ = \{i : r_i = s_i \neq 0\}, \quad E_- = \{i : r_i = -s_i \neq 0\}.$$

Apply Lemma 2.1 to $r + s$. Its magnitude-two layer, divided by two, is the restriction $r \mathbf{1}_{E_+}$, so

$$\sum_{i \in E_+} r_i a_i = 0.$$

Apply the same lemma to $r - s$ to obtain

$$\sum_{i \in E_-} r_i a_i = 0.$$

If the supports of r and s meet, one of E_+ and E_- is nonempty. Minimality of r forces that nonempty set to be all of $\text{supp } r$. Since E_+ and E_- are disjoint, exactly one is all of $\text{supp } r$. Hence $s = r$ on $\text{supp } r$ or $s = -r$ there.

If s is also minimal, this gives $\text{supp } r \subseteq \text{supp } s$. A strict inclusion would make r a signed relation properly supported inside s , contradicting minimality of s . The supports are equal and $s = \pm r$. $\square$

Choose one representative

$$r^{(1)}, \dots, r^{(h)}$$

from every opposite pair of minimal signed relations, and write $B_j = \text{supp } r^{(j)}$. The B_j are pairwise disjoint.

Lemma 2.3 (Every signed relation is a block sum). *Every signed relation q has a unique expression*

$$q = \sum_{j=1}^h \alpha_j r^{(j)}, \quad \alpha_j \in \{-1, 0, 1\}.$$

Proof. For existence, if $q \neq 0$, choose a signed relation u of inclusion-minimal support among those supported in $\text{supp } q$. It is globally minimal. The first paragraph of the proof of Lemma 2.2 says that q agrees with either u or $-u$ on all of $\text{supp } u$. Subtract that agreeing copy. The remaining vector is zero or a signed relation, is zero on $\text{supp } u$, and, when nonzero, has strictly smaller support. Induction on support size proves existence. Uniqueness follows coordinatewise because the block supports are disjoint and every coordinate on a block is nonzero in its block vector. $\square$

Lemma 2.4 (Complete short-kernel classification). *If $c \in \{-2, -1, 0, 1, 2\}^n$ and $\sum_i c_i a_i = 0$, then uniquely*

$$c = \sum_{j=1}^h t_j r^{(j)}, \quad t_j \in \{-2, -1, 0, 1, 2\}.$$

Conversely, every such block sum is a relation in $\{-2, -1, 0, 1, 2\}^n$.

Proof. Retain the two layers rather than merging them. Define

$$u_i = \begin{cases} c_i, & |c_i| = 1, \\ 0, & |c_i| \neq 1, \end{cases} \quad v_i = \begin{cases} c_i/2, & |c_i| = 2, \\ 0, & |c_i| \neq 2. \end{cases}$$

Then $c = u + 2v$ and $\text{supp } u \cap \text{supp } v = \emptyset$. Lemma 2.1 says that each nonzero layer among u and v is a signed relation. A zero layer has the unique empty (all-zero) block decomposition. Applying Lemma 2.3 to each nonzero layer, and using that convention for a zero layer, gives

$$u = \sum_j \alpha_j r^{(j)}, \quad v = \sum_j \beta_j r^{(j)}, \quad \alpha_j, \beta_j \in \{-1, 0, 1\}.$$

If $\alpha_j \neq 0$, the entire block B_j lies in $\text{supp } u$; if $\beta_j \neq 0$, the entire block lies in $\text{supp } v$. Their supports are disjoint, so $\alpha_j \beta_j = 0$ for every j . Hence

$$t_j = \alpha_j + 2\beta_j \in \{-2, -1, 0, 1, 2\},$$

which proves existence. Disjoint block supports give uniqueness. Conversely, each block vector is a relation, and disjointness ensures that every coordinate of $\sum_j t_j r^{(j)}$ remains in the stated coefficient set. $\square$

Every block has size at least three. A one-coordinate signed relation would force some positive a_i to be zero. A two-coordinate signed relation would either add two positive integers to zero or force two distinct elements of A to be equal. Both are impossible.

3 Counting ternary collision classes

Define

$$\Phi : \{0, 1, 2\}^n \longrightarrow \mathbb{Z}, \quad \Phi(x) = \sum_{i=1}^n x_i a_i,$$

and let $T(A)$ be its image. By Lemma 2.4,

$$\Phi(x) = \Phi(y) \iff x - y = \sum_j t_j r^{(j)}$$

for unique scalars $t_j \in \{-2, -1, 0, 1, 2\}$.

Fix a block B_j of size m . On the block cube $\{0, 1, 2\}^{B_j}$, reverse the coordinate on every position at which $r_i^{(j)} = -1$ by the exact cube involution

$$(F_j x)_i = \begin{cases} x_i, & r_i^{(j)} = 1, \\ 2 - x_i, & r_i^{(j)} = -1. \end{cases}$$

Then

$$x - y = t r^{(j)}|_{B_j} \iff F_j x - F_j y = t(1, \dots, 1).$$

Each diagonal class in $\{0, 1, 2\}^m$ has exactly one representative with minimum coordinate zero: subtract the minimum coordinate. There are 3^m ternary vectors, and exactly 2^m have all coordinates in $\{1, 2\}$. Thus the block contributes exactly

$$3^m - 2^m$$

collision classes. If t coordinates lie outside the union of all blocks and $m_j = |B_j|$, disjointness makes the quotient a direct product and gives the exact count

$$|T(A)| = 3^t \prod_{j=1}^h (3^{m_j} - 2^{m_j}), \quad t + \sum_{j=1}^h m_j = n. \quad (1)$$

For $m \geq 3$, put $f_m = 3^m - 2^m$. We have $f_3 = 19$ and

$$f_{m+1} = 3f_m + 2^m > 3f_m.$$

Since $3 > 19^{1/3}$, induction from $m = 3$ yields

$$3^m - 2^m \geq 19^{m/3} \quad (m \geq 3).$$

Also $3^t \geq 19^{t/3}$. Equation (1) therefore gives

$$|T(A)| \geq 19^{n/3}.$$

Every ternary sum is an integer in the interval

$$0 \leq \Phi(x) \leq 2 \sum_i a_i \leq 2nN.$$

Consequently

$$19^{n/3} \leq |T(A)| \leq 2nN + 1,$$

and hence

$$g_4(n) \geq \frac{19^{n/3} - 1}{2n}. \quad (2)$$

Remark 3.1 (Finite strengthening). Because the generators are distinct,

$$\sum_i a_i \leq nN - \frac{n(n-1)}{2}.$$

The same proof gives the independent finite refinement

$$g_4(n) \geq \left\lceil \frac{19^{n/3} + n(n-1) - 1}{2n} \right\rceil.$$

This refinement is not needed for the exponential rate.

4 The seven-digit construction modulo 19

Let

$$L(u, v, w) = u + 7v + 8w$$

and let

$$D = L(\{0, 1\}^3) = \{0, 1, 7, 8, 9, 15, 16\} \subset \mathbb{Z}/19\mathbb{Z}.$$

Whenever its elements are used as base-19 digits or in integer equalities, D denotes these displayed canonical representatives in $\{0, 1, \dots, 18\} \subset \mathbb{Z}$; reduction modulo 19 is stated explicitly. The residue 8 has the two binary representatives $(0, 0, 1)$ and $(1, 1, 0)$; this collision is retained throughout the argument.

Lemma 4.1 (Bounded modular kernel). *For $u, v, w \in \{-2, -1, 0, 1, 2\}$,*

$$u + 7v + 8w \equiv 0 \pmod{19} \iff (u, v, w) = t(1, 1, -1)$$

for a unique $t \in \{-2, -1, 0, 1, 2\}$.

Proof. The reverse implication is immediate from $1 + 7 - 8 = 0$. For the forward implication put

$$p = u + w, \quad q = v + w.$$

Then $|p|, |q| \leq 4$ and $u + 7v + 8w = p + 7q$. Its absolute value is at most 32, so divisibility by 19 forces $p + 7q \in \{-19, 0, 19\}$.

If $p + 7q = 19$, the bounds $|p|, |q| \leq 4$ force $(p, q) = (-2, 3)$. But $v + w = 3$ forces $w \geq 1$, so $u + w \geq -1$, contradicting $p = -2$. Negation excludes -19 . Thus $p + 7q = 0$. The bound $|p| \leq 4$ forces $q = 0$, then $p = 0$, and therefore $u = v = -w$. $\square$

Proposition 4.2 (Digit rigidity). *The set D contains no nonconstant four-term arithmetic progression modulo 19.*

Proof. Suppose $z_0, z_1, z_2, z_3 \in D$ have vanishing consecutive second differences modulo 19. Choose arbitrary binary representatives $x_i \in \{0, 1\}^3$ with $L(x_i) = z_i$. Both vectors

$$x_0 - 2x_1 + x_2, \quad x_1 - 2x_2 + x_3$$

belong to $\{-2, -1, 0, 1, 2\}^3$ and lie in the modular kernel of L . Lemma 4.1 makes each a scalar multiple of $(1, 1, -1)$.

Now retain the exact projection

$$\pi : \mathbb{Z}^3 \longrightarrow \mathbb{Z}^2, \quad \pi(a, b, c) = (a + c, b + c).$$

Its kernel contains $(1, 1, -1)$, and

$$L(a, b, c) = \pi_1(a, b, c) + 7\pi_2(a, b, c).$$

Applying π shows that $\pi(x_0), \pi(x_1), \pi(x_2), \pi(x_3)$ is an ordinary integer four-term progression in $\{0, 1, 2\}^2$. In either coordinate its first and fourth values differ by three times the integer common difference, but both lie in $[0, 2]$. The common difference is therefore zero in each coordinate. All four projections, and hence all four L -values, are equal. Thus the original modular progression is constant. Either representative of the duplicated digit 8 gives the same projection $(1, 1)$, so no representative choice has been suppressed. $\square$

5 Carry-free base-19 lifting

For $m \geq 1$, define the set of $3m$ distinct positive generators

$$A_m = \bigcup_{j=0}^{m-1} 19^j \{1, 7, 8\}.$$

Each subset chooses a binary triple in each block. Hence its contribution at base-19 position j is a digit in D . Conversely, every digit of D is a subset sum of $\{1, 7, 8\}$. Since the largest digit is $16 < 19$, no carry occurs, and therefore

$$H(A_m) = \left\{ \sum_{j=0}^{m-1} d_j 19^j : d_j \in D \right\}. \quad (3)$$

Proposition 5.1 (Least-differing-digit lift). *For every $m \geq 1$, the set $H(A_m)$ contains no nonconstant four-term arithmetic progression over the integers.*

Proof. Suppose

$$y_i = y_0 + i\delta \in H(A_m) \quad (0 \leq i \leq 3)$$

for a nonzero integer δ . Let $r = v_{19}(|\delta|)$, so $\delta = 19^r e$ with $19 \nmid e$. The exponent satisfies $r < m$: indeed, every element of $H(A_m)$ lies in

$$\left[0, 16 \sum_{j=0}^{m-1} 19^j\right] = \left[0, \frac{8}{9}(19^m - 1)\right] \subset [0, 19^m),$$

so a nonzero difference of two such elements cannot be divisible by 19^m .

The four numbers are congruent modulo 19^r . By uniqueness of the standard base-19 expansion in (3), their lower r digits are identical. Let c be the common integer represented by those digits and put

$$z_i = \frac{y_i - c}{19^r}.$$

Each z_i still has all base-19 digits in D , and the four z_i form an integer progression of common difference e . Reducing modulo 19 now reads their least remaining digits and produces a four-term progression in D with nonzero common difference $e \bmod 19$. This contradicts Proposition 4.2. $\square$

The generators in A_m are distinct because their 19-adic valuations give the block index and the multipliers 1, 7, 8 are distinct within a block. Thus

$$|A_m| = 3m, \quad \max A_m = 8 \cdot 19^{m-1},$$

and Proposition 5.1 gives

$$g_4(3m) \leq 8 \cdot 19^{m-1}.$$

For arbitrary $n \geq 1$, take $m = \lceil n/3 \rceil$ and delete any $3m - n$ generators. If $B \subseteq A_m$, then $H(B) \subseteq H(A_m)$; deletion cannot create a progression. Therefore

$$g_4(n) \leq 8 \cdot 19^{\lceil n/3 \rceil - 1}. \quad (4)$$

6 Completion of the proof

Equations (2) and (4) prove the finite bounds in Theorem 1.1. For the asymptotic statement, the n th root of the lower expression is

$$19^{1/3}(1 - 19^{-n/3})^{1/n}(2n)^{-1/n},$$

which tends to $19^{1/3}$, while the n th root of the upper expression is

$$19^{\lceil n/3 \rceil / n} (8/19)^{1/n},$$

which has the same limit. The squeeze theorem proves

$$\lim_{n \rightarrow \infty} g_4(n)^{1/n} = 19^{1/3}.$$

7 Corrections and exact verification status

The source post loses superscripts in two plain-text phrases: “ $2m$ vectors” must be 2^m vectors, and “ $19r$ ” must be 19^r . It also compresses the disjoint-layer sentence in Lemma 2.4, the common-low-digit removal in Proposition 5.1, and the fact that $r < m$. Those steps are supplied above. Finally, the statement that a three-generator block has nineteen ternary classes requires the ambient four-progression-free hypothesis and the complete short-kernel classification. It is not a statement about an arbitrary triple with one visible signed relation.

The accompanying Lean 4 module certifies, without ‘sorry’, the following exact claims:

- (i) the bounded kernel in Lemma 4.1;
- (ii) the two-second-difference rigidity of the seven-digit set;
- (iii) the recursive base-19 lift for every word length;
- (iv) the identification of each binary block choice with one of the seven digits;
- (v) the relation-splitting algebra in Lemma 2.1.

The Lean module does not yet certify the full minimal-block decomposition, the global product cardinality in (1), or the real-asymptotic squeeze. Those parts are proved in ordinary mathematics in this note and are not mislabeled as Lean-complete.

The executable certificate independently enumerates all 125 bounded coefficient triples, all 342 start/nonzero-step modular progressions, all seven-digit second-difference quadruples, the lifted languages through four blocks, and the exact deletion choices through nine generators. A separate stress test checks the minimal-block and ternary-count statements over a stated finite search domain. Finite enumeration is corroborating evidence, not a replacement for the general proofs.

Novelty status. The checked sources below do not state the exact $19^{1/3}$ rate, but this is not a priority claim. A bounded literature search cannot establish novelty. The broad catalogue problem asks for estimates across $k \geq 3$ [3]. Costa’s negative answer for the historical $k = 3$ lower-bound question and the present $k = 4$ exponential-rate theorem concern different subproblems and do not exhaust that broader request. The present theorem may be new, overlooked, independently rediscovered, or in need of further community review.

References

- [1] Anonymous/deleted Reddit contributor. A proposed solution for the $k = 4$ exponential-rate case of erdős problem 817. Post to r/LLMmathematics, 2026. Account subsequently deleted; https://www.reddit.com/r/LLMmathematics/comments/1wed5gb/a_proposed_solution_for_the_k4_exponentialrate/; accessed 12 September 2026.
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- [6] Samuel KORSKY. Arithmetic progression-free subset-sum sets, 2026. Version 1, 23 June 2026; <https://arxiv.org/abs/2606.24139>.